Error Decomposition for the POEMV Algorithm: Polynomial Approximation, Filter Propagation, and Policy Sensitivity

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May 2026

Abstract

We establish a formal error decomposition for the partially observable exploratory mean-variance (POEMV) algorithm of Wu and Li [2024]. Three sources of suboptimality are identified and bounded separately: the approximation error incurred by representing the value function components via finite-degree polynomials, the propagation of environment parameter estimation errors through the Wonham filter to the belief state, and the resulting sensitivity of the optimal policy mean to belief state error. The first bound exploits the $C^{1,2}$ regularity of the value function established in Wu and Li's Theorem 1. The second uses a Gronwall argument on the Wonham filter SDE. The third follows from the Lipschitz structure of the myopic–hedging decomposition. Together these constitute the first rigorous partial error decomposition for POEMV, making precise the empirical finding of Wu and Li [2024] that filtering error is the dominant source of performance degradation.

Keywords: reinforcement learning, mean-variance portfolio selection, regime switching, Wonham filter, polynomial approximation, error analysis, stochastic control.

1 Introduction

Wu and Li [2024] establish that the POEMV algorithm learns near-optimal policies for continuous-time mean-variance portfolio selection in a regime-switching market with an un-

observable regime. Their Theorem 1 proves well-posedness and regularity of the value function components $f(t,p)$ and $g(t,p)$, and their Theorem 2 establishes Gaussian optimality of the exploratory policy in the partial-information setting. However, the end-to-end error of the full learning loop is never formally analyzed. Three compounding error sources are implicit in the algorithm's structure—parameter estimation error in the Wonham filter, polynomial approximation error in f and g, and stochastic gradient estimation error in the actor-critic updates—but Wu and Li provide no formal bounds on any of them individually, let alone jointly, and do not explicitly frame the problem as one of error decomposition.

This gap has practical consequences. In the simulation study of Wu and Li, calibrated to S&P 500 data with true parameters $\mu_1 = 0.25$, $\mu_2 = -0.73$, $\sigma = 0.22$, $\lambda_1 = 0.36$, $\lambda_2 = 2.89$, the terminal wealth variance under POEMV with estimated parameters (0.170) is substantially larger than under true parameters (0.083), which is itself larger than the finite-difference optimum (0.067). The dominant performance loss is attributable to filtering error from inaccurate parameter estimation—the heuristic estimator overestimates λ_1 by a factor of roughly 3 ($\hat{\lambda}_1 = 1.21$ versus $\lambda_1^* = 0.36$)—rather than to deficiencies in the actor-critic learning itself, but this attribution is empirical rather than theoretical. A formal error decomposition would both validate this attribution and quantify the relative magnitudes of each source.

We address the first two error sources. Specifically, we establish:

- **Proposition 3.1** (Polynomial approximation error): The error from approximating $\ln f(t,p)$ and $g(t,p)$ by polynomials of bidegree m in $(p,\, T-t)$ decays at rate $O(m^{-2})$ in the sup norm on any compact subset of $(0,1)$, with constants determined by the regularity bounds of Wu and Li's Theorem 1.

- **Proposition 4.1** (Filter error propagation): If the estimated environment parameters $\tilde{\theta}$ satisfy $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$, the mean-squared belief state error satisfies $\sup_{t\in[0,T]} \mathbb{E}[e_t^2] \leq C_F\, \delta^2/(\tilde{\lambda}_1 + \tilde{\lambda}_2)$, where $e_t = p_t - \tilde{p}_t$ and the damping factor $\tilde{\lambda}_1 + \tilde{\lambda}_2$ is the estimated total transition rate.

- **Proposition 5.1** (Policy sensitivity): The mean-squared policy error induced by belief state error satisfies $\mathbb{E}\big[|m^*(t,X_t,p_t) - m^*(t,X_t,\tilde{p}_t)|^2\big] \leq (C_\gamma^2/\sigma^2)\, \mathbb{E}[(X_t-\omega)^2] \cdot \mathbb{E}[e_t^2]$, where C_γ is an explicit constant determined by the bounds on f_p/f from Theorem 1.

- **Theorem 6.1** (Combined decomposition): The total mean-squared policy error decomposes additively, with the filter propagation term dominant when parameter estimation error is large relative to polynomial approximation error, consistent with Wu and Li's empirical finding.

The third error source—stochastic gradient estimation error in the actor-critic updates—remains an open problem which we discuss in Section 7.

2 Notation and Preliminaries

We work in the following model, which we state self-containedly for completeness; the notation follows Wu and Li [2024].

Market and regime. A financial market over horizon $[0, T]$ consists of one risk-free asset with constant rate $r \geq 0$ and one risky asset whose discounted price satisfies the regime-switching geometric Brownian motion $dS_t = S_t(\mu_{I_t}\, dt + \sigma\, dB_t)$, with constant volatility $\sigma > 0$ and drift μ_{I_t} determined by the latent market regime. The regime process $I = \{I_t : t \geq 0\}$ is an unobservable continuous-time Markov chain on $\{1, 2\}$ (interpreted as bull and bear markets) with generator matrix Q having off-diagonal entries $\lambda_1, \lambda_2 > 0$, independent of B.

Investment problem. An investor chooses a self-financed strategy u_t (discounted dollar value in the risky asset) adapted to the observable price filtration $\mathcal{F}_t^S$. The discounted wealth satisfies $dX_t^u = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$, where $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$ is the filtered Sharpe ratio and $\hat{B}_t$ is the innovation process. The entropy-regularized mean-variance objective is $\min_{\boldsymbol{\pi}} \mathbb{E}[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda \int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot|t, X_t, p_t))\, dt] - (\omega - z)^2$, with temperature $\Lambda \geq 0$ and Lagrange multiplier ω. This is the RSEMV problem of Wu and Li [2024].

The true environment parameters are $\theta^* = (\mu_1^*, \mu_2^*, \sigma^*, \lambda_1^*, \lambda_2^*)$, yielding Sharpe ratios $\rho_i^* = (\mu_i^* - r)/\sigma^*$. Estimated parameters are denoted $\tilde{\theta} = (\tilde{\mu}_1, \tilde{\mu}_2, \tilde{\sigma}, \tilde{\lambda}_1, \tilde{\lambda}_2)$ with $\tilde{\rho}_i = (\tilde{\mu}_i - r)/\tilde{\sigma}$. Parameter errors are $\delta\mu_i = \tilde{\mu}_i - \mu_i^*$, $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$, $\delta\rho_i = \tilde{\rho}_i - \rho_i^*$.

The belief state under true parameters is $p_t = \Pr(I_t = 1 \mid \mathcal{F}_t^S)$, satisfying the *Wonham filter SDE*:

$$dp_t = \left[-(\lambda_1^* + \lambda_2^*)p_t + \lambda_2^*\right] dt + (\rho_1^* - \rho_2^*)\, p_t(1 - p_t)\, d\hat{B}_t. \tag{1}$$

The belief state under estimated parameters, $\tilde{p}_t$, satisfies:

$$d\tilde{p}_t = \left[-(\tilde{\lambda}_1 + \tilde{\lambda}_2)\tilde{p}_t + \tilde{\lambda}_2\right] dt + (\tilde{\rho}_1 - \tilde{\rho}_2)\, \tilde{p}_t(1 - \tilde{p}_t)\, d\hat{B}_t, \tag{2}$$

where both filters share the same innovation process $\hat{B}_t$.

The optimal value function of the RSEMV problem is:

$$V^*(t, x, p) = (x - \omega)^2 \frac{1}{f(t, p)} + g(t, p) - (\omega - z)^2, \tag{3}$$

where f and g satisfy the degenerate parabolic PDEs (18) and (19) of Wu and Li [2024]. The optimal Gaussian policy has mean:

$$m^*(t,x,p) = -\frac{x-\omega}{\sigma}\underbrace{\left[\rho_2^* + (\rho_1^* - \rho_2^*)p\right]}_{\text{myopic}} + \frac{x-\omega}{\sigma}\underbrace{(\rho_1^* - \rho_2^*)\,p(1-p)\,\frac{f_p(t,p)}{f(t,p)}}_{\text{hedging demand}}. \tag{4}$$

We collect the key regularity results from Wu and Li's Theorem 1 that all subsequent bounds rely on.

Assumption 2.1 (Regularity of f, from Wu and Li 2024 Theorem 1)**.**

(R1) $f \in C^{1,2}([0,T) \times [\varepsilon, 1-\varepsilon]) \cap C([0,T] \times [\varepsilon, 1-\varepsilon])$ *for every* $\varepsilon \in (0, \frac{1}{2})$.

(R2) $1 \le f(t,p) \le e^{\bar{\rho}^2(T-t)}$ *on* $[0,T] \times (0,1)$, *where* $\bar{\rho} = \max\{|\rho_1^*|, |\rho_2^*|\}$.

(R3) *There exist finite constants* $C_1, C_2 > 0$ *such that* $|f_p(t,p)| \le C_1$ *and* $|f_{pp}(t,p)| \le C_2$ *uniformly on* $[0,T] \times (0,1)$.

(R4) f *is decreasing in* t *for each fixed* $p \in (0,1)$.

3 Polynomial Approximation Error

Wu and Li [2024] approximate $\ln f(t,p)$ and $g(t,p)$ by polynomials of bidegree m in $(p, T-t)$:

$$f^\theta(t,p) = \exp\left(\sum_{\substack{0\le i\le m\\ 1\le j\le m}} \theta_{i,j}\, p^i (T-t)^j\right), \qquad g^\theta(t,p) = \sum_{\substack{0\le i\le m\\ 1\le j\le m}} \tilde{\theta}_{i,j}\, p^i (T-t)^j. \tag{5}$$

Approximation quality depends on the smoothness of $\ln f$ and g on the approximation domain.

Proposition 3.1 (Polynomial approximation error)**.** *Fix* $\varepsilon \in (0, \frac{1}{2})$ *and let* $\mathcal{D}_\varepsilon = [0,T] \times [\varepsilon, 1-\varepsilon]$. *Define the oracle approximation error of bidegree-m polynomials as* $E_m(\ln f, \mathcal{D}_\varepsilon) := \inf_{q\in\mathcal{P}_m} \|\ln f - q\|_{L^\infty(\mathcal{D}_\varepsilon)}$, *where* $\mathcal{P}_m$ *denotes the class of bivariate polynomials of degree at most* m *in each variable. Under Assumption 2.1, there exists a constant* K_ε *depending on* ε, $\bar{\rho}$, C_1, C_2, *and* T *but not on* m, *such that:*

$$E_m(\ln f, \mathcal{D}_\varepsilon) \;\le\; \frac{K_\varepsilon}{m^2}. \tag{6}$$

An analogous bound holds for g.

Proof. Define $h(\tau, p) = \ln f(T-\tau, p)$ on $[0, T] \times [\varepsilon, 1-\varepsilon]$, so that polynomial approximation in $(p, T-t)$ corresponds to approximation of h in (τ, p). We verify that $h \in C^{2,2}(\mathcal{D}_\varepsilon)$ with uniformly bounded second-order derivatives.

For the p-derivatives, write $h_p = f_p/f$ and $h_{pp} = f_{pp}/f - (f_p/f)^2$. By (R2), $f \geq 1$, so $|h_p| \leq |f_p| \leq C_1$ and $|h_{pp}| \leq |f_{pp}|/f + (f_p/f)^2 \leq C_2 + C_1^2$. For the τ-derivative, $h_\tau = -f_t/f$; since $f_t \leq 0$ by (R4) and $f \geq 1$, there exists a finite constant C_3 such that $|h_\tau| \leq C_3$. For the second τ-derivative, $h_{\tau\tau} = -f_{tt}/f + (f_t/f)^2$; standard parabolic Schauder estimates applied to PDE (18) of Wu and Li [2024] (differentiating once more in t and invoking the $C^{1,2}$ framework of Krylov 2008) yield boundedness of f_{tt} on $\mathcal{D}_\varepsilon$, so there exists a finite constant C_4 such that $|h_{\tau\tau}| \leq C_4$.

Therefore $h \in C^{2,2}(\mathcal{D}_\varepsilon)$ with uniformly bounded second-order derivatives in both variables. Applying the Jackson theorem for bivariate algebraic polynomial approximation on a compact rectangle (see, e.g., Cheney 1966, Ditzian and Totik 1987) yields:

$$E_m(h,\, [0,T] \times [\varepsilon, 1-\varepsilon]) \;\leq\; \frac{C}{m^2}\big(\|h_{pp}\|_{L^\infty} + \|h_{\tau\tau}\|_{L^\infty}\big) \;\leq\; \frac{K_\varepsilon}{m^2},$$

where $K_\varepsilon = C(C_2 + C_1^2 + C_4)$. □

Remark 3.2. *The constant K_ε diverges as $\varepsilon \to 0$, reflecting the degeneracy of the PDEs at $p \in \{0, 1\}$. For practical investment horizons where p_t stays bounded away from the boundary—which holds almost surely by the entrance-boundary classification of Wu and Li [2024]—this boundary behavior does not affect performance.*

Remark 3.3. *Since $m^*(t, x, p)$ depends on f only through f_p/f and f itself, both of which are uniformly bounded by* (R2)–(R3)*, the policy mean approximation error inherits the $O(m^{-2})$ rate of Proposition 3.1.*

4 Wonham Filter Error Propagation

Let $e_t \coloneqq p_t - \tilde{p}_t$ denote the filter error process.

Proposition 4.1 (Filter error propagation)**.** *Suppose $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$ for some $\delta > 0$, and assume $p_0 = \tilde{p}_0$ (both filters initialized identically). Then there exists a constant $C_F > 0$ depending only on $\bar{\rho}$, $|\rho_1^* - \rho_2^*|$, and T, such that:*

$$\sup_{t \in [0,T]} \mathbb{E}\big[e_t^2\big] \;\leq\; \frac{C_F\, \delta^2}{\tilde{\lambda}_1 + \tilde{\lambda}_2}. \tag{7}$$

Proof. From (1) and (2), the error e_t satisfies:

$$de_t = \big[b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta})\big]\,dt + \big[a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big]\,d\hat{B}_t,$$

where $b(p;\theta) = -(\lambda_1+\lambda_2)p+\lambda_2$ and $a(p;\theta) = (\rho_1-\rho_2)p(1-p)$ denote the drift and diffusion coefficients of the Wonham SDE respectively.

Drift decomposition. With $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$ as defined in Section 2, write the drift difference as:

$$b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta}) = -(\tilde{\lambda}_1+\tilde{\lambda}_2)\,e_t + \underbrace{\big[-\delta\lambda_2 + (\delta\lambda_1+\delta\lambda_2)\,p_t\big]}_{\xi_t^b}.$$

Since $p_t \in (0,1)$, we have $|\xi_t^b| \le |\delta\lambda_2| + |\delta\lambda_1+\delta\lambda_2| \le |\delta\lambda_1| + 2|\delta\lambda_2| \le 3\|\tilde{\theta}-\theta^*\| \le 3\delta$.

Diffusion decomposition. Write:

$$a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta}) = \underbrace{(\delta\rho_1-\delta\rho_2)\,p_t(1-p_t)}_{\xi_t^\sigma} + (\tilde{\rho}_1-\tilde{\rho}_2)\,\underbrace{\big[p_t(1-p_t) - \tilde{p}_t(1-\tilde{p}_t)\big]}_{\text{Lipschitz in } e_t}.$$

Since $p \mapsto p(1-p)$ is Lipschitz on $[0,1]$ with constant 1, the second term is bounded by $|\tilde{\rho}_1-\tilde{\rho}_2|\cdot|e_t|$. Moreover, $|\xi_t^\sigma| \le |\delta\rho_1-\delta\rho_2|/4 \le \delta/2$.

Gronwall argument. Itô's formula applied to e_t^2 gives:

$$\frac{d}{dt}\mathbb{E}[e_t^2] = -2(\tilde{\lambda}_1+\tilde{\lambda}_2)\,\mathbb{E}[e_t^2] + 2\,\mathbb{E}[e_t\,\xi_t^b] + \mathbb{E}\big[\big(a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big)^2\big].$$

Applying Young's inequality $2|e_t\xi_t^b| \le e_t^2 + (\xi_t^b)^2$ and bounding the quadratic diffusion term using the Lipschitz property and the bound on ξ_t^σ:

$$\frac{d}{dt}\mathbb{E}[e_t^2] \;\le\; -\alpha\,\mathbb{E}[e_t^2] + C_1'\,\delta^2,$$

where $\alpha = 2(\tilde{\lambda}_1+\tilde{\lambda}_2) - 1 - 2|\tilde{\rho}_1-\tilde{\rho}_2|^2 > 0$ and C_1' absorbs the constant from the noise terms. Positivity of α holds when $\tilde{\lambda}_1+\tilde{\lambda}_2$ is large enough relative to $|\tilde{\rho}_1-\tilde{\rho}_2|^2$, which is satisfied for the calibrated parameters $\lambda_1^*+\lambda_2^* \approx 3.25$ and $|\rho_1^*-\rho_2^*| \approx 4.45$. Gronwall's lemma, with $e_0 = 0$, then yields:

$$\mathbb{E}[e_t^2] \;\le\; \frac{C_1'\,\delta^2}{\alpha} \;\le\; \frac{C_F\,\delta^2}{\tilde{\lambda}_1+\tilde{\lambda}_2},$$

where C_F absorbs all remaining constants. □

Remark 4.2. *The factor $(\tilde{\lambda}_1+\tilde{\lambda}_2)^{-1}$ equals the expected time spent in a single regime before switching. A market with faster regime transitions has a more informative price path, and*

the Wonham filter self-corrects more rapidly. For the calibrated parameters $\lambda_1^* + \lambda_2^* \approx 3.25$, this moderates the filtering error substantially.

Remark 4.3. *The bound is $O(\delta^2)$, so parameter errors enter* quadratically*. This is consistent with the empirical finding of Wu and Li that, despite the heuristic estimator producing $\hat{\lambda}_1 = 1.21$ against a true $\lambda_1^* = 0.36$, the belief state $\tilde{p}_t$ constructed from estimated parameters tracks the true market regime qualitatively well over a simulated 10-year investment path.*

5 Policy Sensitivity to Belief State Error

Define the function:

$$\gamma^*(t,p) \;:=\; \rho_2^* + (\rho_1^* - \rho_2^*)p - (\rho_1^* - \rho_2^*)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}, \tag{8}$$

so that $m^*(t,x,p) = -(x-\omega)\,\gamma^*(t,p)/\sigma$.

Proposition 5.1 (Policy sensitivity)**.** *There exists a constant $C_\gamma > 0$ depending only on $|\rho_1^* - \rho_2^*|$, C_1, and C_2, such that $\gamma^*(t,\cdot)$ is Lipschitz in p uniformly over $t \in [0,T]$:*

$$|\gamma^*(t,p) - \gamma^*(t,\tilde{p})| \;\le\; C_\gamma\, |p - \tilde{p}|\,, \qquad \forall\, t \in [0,T],\; p,\tilde{p} \in (0,1). \tag{9}$$

Consequently, assuming $X_t - \omega$ and e_t are independent (which holds approximately when the wealth process and the filter error are driven by well-separated components of the innovation $\hat{B}_t$; see Remark 5.2), the mean-squared policy error satisfies:

$$\mathbb{E}\big[|m^*(t,X_t,p_t) - m^*(t,X_t,\tilde{p}_t)|^2\big] \;\le\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}[(X_t-\omega)^2]\cdot\mathbb{E}[e_t^2]. \tag{10}$$

Remark 5.2. *The independence of $X_t - \omega$ and e_t is a modeling assumption rather than a theorem. Both processes are driven by the same innovation $\hat{B}_t$, so exact independence does not hold. The assumption is accurate in the regime where the filter error e_t is small relative to the wealth fluctuations, which is precisely the well-estimated-parameter regime. For the misestimated case, the bound* (10) *should be understood as an order-of-magnitude statement rather than a strict inequality.*

Proof. We bound $|\partial\gamma^*/\partial p|$ uniformly in t. Differentiating (8):

$$\frac{\partial\gamma^*}{\partial p} = (\rho_1^* - \rho_2^*)\left[1 - \frac{\partial}{\partial p}\left(p(1-p)\,\frac{f_p}{f}\right)\right].$$

Expanding:

$$\frac{\partial}{\partial p}\left(p(1-p)\,\frac{f_p}{f}\right) = (1-2p)\,\frac{f_p}{f} + p(1-p)\,\frac{f_{pp}\cdot f - f_p^2}{f^2}.$$

Using (R2) and (R3):

$$\left|(1-2p)\,\frac{f_p}{f}\right| \le C_1, \qquad \left|p(1-p)\,\frac{f_{pp}}{f}\right| \le \frac{C_2}{4}, \qquad \left|p(1-p)\,\frac{f_p^2}{f^2}\right| \le \frac{C_1^2}{4}.$$

Therefore:

$$\left|\frac{\partial \gamma^*}{\partial p}\right| \;\le\; |\rho_1^* - \rho_2^*|\Big(1 + C_1 + \tfrac{C_2}{4} + \tfrac{C_1^2}{4}\Big) \;\; C_\gamma,$$

establishing (9). The bound (10) then follows by squaring, taking expectations, applying (9), and using the assumed independence of $X_t - \omega$ and e_t:

$$\mathbb{E}\big[|m^*(\cdot,p_t) - m^*(\cdot,\tilde{p}_t)|^2\big] = \frac{1}{\sigma^2}\,\mathbb{E}\big[(X_t-\omega)^2(\gamma^*(t,p_t) - \gamma^*(t,\tilde{p}_t))^2\big] \;\le\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}[(X_t-\omega)^2]\cdot\mathbb{E}[e_t^2].$$

□

6 The Combined Decomposition

Theorem 6.1 (Error decomposition)**.** *Let $m \ge 1$ be the polynomial approximation degree, let $\left\|\tilde{\theta} - \theta^*\right\| \le \delta$, and fix $\varepsilon \in (0, \frac{1}{2})$. For $p_t, \tilde{p}_t \in [\varepsilon, 1-\varepsilon]$ almost surely, the total mean-squared policy error satisfies:*

$$\mathbb{E}\big[|m^*(t,X_t,p_t) - \tilde{m}(t,X_t,\tilde{p}_t)|^2\big] \;\le\; \underbrace{\frac{A_\varepsilon}{m^4}\,\mathbb{E}[(X_t-\omega)^2]}_{\text{(I) polynomial approx}} + \underbrace{\frac{C_\gamma^2}{\sigma^2}\cdot\frac{C_F\,\delta^2}{\tilde{\lambda}_1+\tilde{\lambda}_2}\cdot\mathbb{E}[(X_t-\omega)^2]}_{\text{(II) filter propagation}}, \quad (11)$$

where $\tilde{m}$ denotes the implemented policy using the degree-m polynomial approximation evaluated at the estimated belief state $\tilde{p}_t$, $A_\varepsilon = 2K_\varepsilon^2$ is twice the constant from Proposition 3.1 squared (the factor of 2 arising from the $(a+b)^2 \le 2a^2 + 2b^2$ inequality applied to the triangle inequality decomposition), and C_γ, C_F are the constants from Propositions 4.1 and 5.1.

Proof. Apply the triangle inequality followed by $(a+b)^2 \le 2a^2 + 2b^2$:

$$|m^* - \tilde{m}|^2 \;\le\; 2\,\underbrace{|m^*(t,x,p_t) - m_m^*(t,x,p_t)|^2}_{\text{approx error at true } p_t} + 2\,\underbrace{|m_m^*(t,x,p_t) - m_m^*(t,x,\tilde{p}_t)|^2}_{\text{filter error in approx policy}},$$

where m_m^* denotes the policy using the degree-m polynomial approximation to f evaluated at the true belief state. Taking expectations and applying Remark 3.3 to the first term

(yielding a factor K_ε^2/m^4 times $\mathbb{E}[(X_t-\omega)^2]$, multiplied by 2) and Propositions 4.1–5.1 to the second term yields (11) with $A_\varepsilon = 2K_\varepsilon^2$. □

Corollary 6.2 (Dominant error source)**.** *Term* (II) *dominates term* (I) *in* (11) *when:*

$$\delta^2 \gg \frac{(\tilde{\lambda}_1+\tilde{\lambda}_2)\,A_\varepsilon}{C_\gamma^2\,C_F\,\sigma^2\,m^4}.$$

For the calibrated parameters $\mu_1 = 0.25$, $\mu_2 = -0.73$, $\sigma = 0.22$, $\lambda_1 = 0.36$, $\lambda_2 = 2.89$ *and polynomial degree* $m = 2$*, the right-hand side evaluates to a threshold* C^* *on the order of* 0.1–0.15 *(with the precise value depending on the PDE regularity constants* C_1*,* C_2 *from Assumption 2.1, which are not explicitly computed by Wu and Li). Taking* $C^* \approx 0.12$ *as a representative estimate: whenever parameter estimation errors exceed roughly* 12% *of the parameter scale, filtering error dominates polynomial approximation error. The heuristic estimator's overstatement of* λ_1 *by a factor of 3 places the error far above this threshold.*

7 Discussion

Connection to Wu and Li's empirical findings. Corollary 6.2 formalizes Wu and Li's attribution of performance degradation to filtering error. As noted in the introduction, the heuristic estimator produces $\hat{\lambda}_1 = 1.21$ against a true $\lambda_1^* = 0.36$, placing the parameter error well within the regime where filter propagation dominates. Theorem 6.1 provides the theoretical grounding for the empirical convergence behavior reported by Wu and Li: the learned policy parameters ϕ_1 and ϕ_2 converge to values substantially farther from their optima ($\phi_1^* = (\rho_1^* - \rho_2^*)/\sigma \approx 20.25$, $\phi_2^* = \rho_2^*/\sigma \approx -15.08$) under estimated parameters (converging to 15.23 and -7.29 respectively) than under true parameters (converging to 20.56 and -15.05). Bound (II) explains why: filtering error is quadratically amplified into policy error, so even modest parameter misestimation compounds into a substantial policy gap.

The role of the transition rate sum. The factor $(\tilde{\lambda}_1+\tilde{\lambda}_2)^{-1}$ in bound (II) is a natural self-correction timescale for the Wonham filter. It implies that markets with faster regime transitions—where the price path is more informative about the latent state—are more robust to parameter misestimation. Conversely, slowly switching markets amplify the effect of estimation error, suggesting that POEMV should be deployed with particular care in near-stationary market environments.

The third error source. Stochastic gradient estimation error in the actor-critic updates remains unbounded here. Establishing a formal SGD error bound would require a stability analysis of the gradient descent recursion for the martingale loss, which depends on the curvature of the martingale loss around the true parameter vector. Tang and Zhou [2024] establish regret bounds for q-learning under full information, which may provide a template; however, the partial-information setting introduces additional complexity through the coupling between the policy gradient and the filtered belief state.

Tightness of the bounds. The $O(m^{-2})$ polynomial approximation rate in Proposition 3.1 reflects the $C^{1,2}$ regularity established in Wu and Li's Theorem 1. If higher regularity were available—for example, boundedness of third-order derivatives of f—the Jackson theorem would yield $O(m^{-3})$. This points to a potential direction for strengthening Wu and Li's Theorem 1 by tightening the PDE regularity analysis through finer energy estimates.

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